higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL CONSTRUCTION N6

2 APRIL 2019

This marking guideline consists of 11 pages.

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QUESTION 1

All references taken from SANS 10100-1 (2000)

Fcu = 20 MPa — Table 2 (4.1.5.1)
Fy = 250 MPa — Table 3 (4.1.5.2)
Span = 4,5 metres — CL 4.3.1.2
Density of reinforced concrete = 2 400 kg/m^3

Steel in tension:

$As = 2 \times \frac{\pi d^2}{4}$ $As = 2 \times \frac{\pi 16^2}{4}$✓ $As = 402$ mm^2✓ (2)

Minimum area of reinforcement:

$\frac{100\ AS}{AC}$ — Table 23

$= \frac{100 \times 402}{200 \times 325}$

$= 0{,}62 > 0{,}45$✓ OK — CL 4 11.5.4 (1)

Maximum area of reinforcement:

4% of AC
4% × (325 × 200) = 2 600 mm^2✓ OK (1)

Distance of lever arm (Z): — CL 4.3.3.4

$Z = d\left\{0{,}5 + \sqrt{0{,}25 - \frac{K^1}{0{,}9}}\right\}$

Use eff d = 400 mm
where cover = 50 mm

$Z = 275\left\{0{,}5 + \sqrt{0{,}25 - \frac{0{,}156}{0{,}9}}\right\}$✓

$Z = 275(0{,}777)$

Z = 213,68 mm✓ (2)

Maximum moment of resistance in tension:

Where: $As = \frac{M}{0{,}87\ fy\ Z}$ CL 4.3.3.4

Then: $M = As \times 0{,}87 \times fy \times Z$

$M = 402\ mm^2 \times 0{,}87 \times 250\ N/mm^2 \times 213{,}68\ mm$✓

$M = 18{,}68\ kNm$✓ (2)

Total load:

Where: Bending moment max = $\frac{wl^2}{8}$ $18{,}68 =$

$\frac{w \times 4{,}5^2}{8}$ ✓ $w = 18{,}68 \times 8/4{,}5^2$

$w = 7{,}4\ kN/m$✓ (2)

Dead load:

Dead load = Vol × density × g.a × 10^{-3} ✓ CL 4.3.3.4

Dead load = $0{,}325 \times 0{,}20 \times 1 \times 2\ 400\ kg/m^3 \times 9{,}81 \times 10^{-3}$

Dead load = $1{,}53\ kN/m$✓ (2)

Max distributed load:

Total load = 1,2 (Gn) + 1,6 (Qn)✓ CL 4.2.2.1

7,4 kN/m = 1,2 (1,53) + 1,6 (Qn)

7,4 kN/m – 1,836 = 1,6 (Qn)✓

Qn = 5,564/1,6

Max distributed load = 3,48 kN/m✓ (3)

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QUESTION 2

2.1 Cross-sectional area of the steel section:

$\text{Area} = 2(180 \times 12) + (9 \times 246)$✓

$\text{Area} = 4\,320 + 2\,214$

$\text{Cross-sectional area} = 6\,534\ \text{mm}^2$✓ (2)

2.2 Second moment of area about the x-x axis:

$$I_{xx} = 2\left(\frac{bd^3}{12} + al^2\right) + \left(\frac{bd^3}{12}\right)$$

$$I_{xx} = 2\left(\frac{180 \times 12^3}{12} + 180 \times 12 \times 135^2\right) + \left(\frac{9 \times 246^3}{12}\right)$$✓✓

$I_{xx} = 2(25\,920\ \text{mm}^4 + 39\,366\,000\ \text{mm}^4) + 11\,165\,202\ \text{mm}^4$

$I_{xx} = 78\,783\,840\ \text{mm}^4 + 11\,165\,202\ \text{mm}^4$✓

$I_{xx} = 89\,949\,042\ \text{mm}^4$✓ (4)

2.3 Second moment of area about the y-y axis:

$$I_{yy} = 2\left(\frac{bd^3}{12}\right) + \left(\frac{bd^3}{12}\right)$$

$$I_{yy} = 2\left(\frac{12 \times 180^3}{12}\right) + \left(\frac{246 \times 9^3}{12}\right)$$✓✓

$I_{yy} = 11\,664\,000\ \text{mm}^4 + 14\,944,5\ \text{mm}^4$

$I_{yy} = 11\,678\,944{,}5\ \text{mm}^4$✓ (3)

2.4 Minimum second moment of area:

$\text{Least second moment of area} = I_{yy} = 11\,678\,944{,}5\ \text{mm}^4$✓ (1)

2.5 Minimum radius of gyration:

$\text{Least second moment of area (I)} = I_{yy} = 11\,678\,944{,}5\ \text{mm}^4$

$$r_{min} = \sqrt{\frac{Iyy}{area}}$$

$$r_{min} = \sqrt{\frac{11\,678\,944,5\ mm^4}{6\,534\ mm^2}}$$✓

$r_{min} = 42{,}28\ \text{mm}$✓ (2)

2.6 Slenderness ratio:

$$L/r = \frac{4\ 650}{42{,}28} = 109{,}98 \ \ (\text{say } 110)\checkmark$$

From Table 17 (SABS 0162-1984):

$110 = 69$ MPa✓ (2)

2.7 Maximum axial load:

$\text{Load} = \text{Stress} \times \text{Area}$

$\text{Load} = 69\ \text{N/mm}^2 \times 6\ 534\ \text{mm}^2$

$\text{Load} = 657{,}85$ kN✓ (1)

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QUESTION 3

All references taken from SANS 10100-1 (2000)

$Fcu = 20$ MPa — Table 2 (4.1.5.1)
$Fy = 250$ MPa — Table 3 (4.1.5.2)
$\text{Width} = 1{,}35$ metres
$\text{Density of reinforced concrete} = 2\ 425\ \text{kg/m}^3$

3.1 Length of the slope:

$\text{Length of slope} = \sqrt{2\ 025^2 + 1\ 600^2}\checkmark$

$\text{Length of slope} = 2\ 580{,}8$ mm $(2{,}585$ m)✓

Load calculations:
Waist:

$W = \text{Vol} \times \text{density} \times \text{g.a} \times 10^{-3}$
$W = 2{,}585 \times 1{,}35 \times 0{,}140 \times 2\ 425\ \text{kg/m}^3 \times 9{,}81 \times 10^{-3}\checkmark$
$W = 11{,}62$ kN✓

Treads or steps:

$W = \text{Vol} \times \text{density} \times \text{g.a} \times 9{,}81 \times 10^{-3}$
$W = ½\text{bh} \times \text{width} \times \text{density} \times \text{g.a} \times 9{,}81 \times 10^{-3}$
$W = ½ \times 1{,}60 \times 0{,}225 \times 1{,}35 \times 2\ 425\ \text{kg/m}^3 \times 9{,}81 \times 10^{-3}\checkmark$
$W = 5{,}781$ kN✓

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Total design dead load:

$1{,}2\ \text{Gn}\ (11{,}62\ \text{kN} + 5{,}781\ \text{kN}) = 20{,}881\ \text{kN}$✓ CL 4.2.2.1

Design imposed load:

$1{,}6\ \text{Qn}\ (4{,}5\ \text{kN/m}^2 \times 1{,}35\ \text{m} \times 2{,}025\ \text{m}) = 19{,}68\ \text{kN}$✓ CL 4.2.2.1 (8)

3.2 Bending moment maximum:

$$\text{BM}_{max} = \frac{WL}{10} + \frac{WL}{10}$$

$$\text{BM}_{max} = \frac{20{,}881 \times 2{,}025}{10} + \frac{19{,}68 \times 2{,}025}{10}$$ ✓✓

$$\text{BM}_{max} = 4{,}23 + 3{,}99$$

$$\text{BM}_{max} = 8{,}46\ \text{kNm}$$✓ (3)

3.3 Value for K:

$$K = \frac{BM}{fcu\ b\ d^2}$$ CL 4.3.3.4.1

Let Eff depth (d)
= 140 – 25 cover
= 115 mm

$$K = \frac{8{,}46 \times 10^6}{20 \times 1\ 350 \times 115^2}$$✓

$K = 0{,}024 < K^1 = 0{,}156$✓

Only tension reinforcement will be required.✓ (3)

3.4 Distance of lever arm (Z):

$$Z = d\left\{0{,}5 + \sqrt{0{,}25 - \frac{K}{0{,}9}}\right\} \leq 0{,}95\text{d}$$ CL 4.3.3.4.1

$$Z = 115\left\{0{,}5 + \sqrt{0{,}25 - \frac{0{,}024}{0{,}9}}\right\} \leq 0{,}95 \times 115$$✓

$Z = 115(0{,}973) \leq 0{,}95 \times 115$

$Z = 111{,}9\ \text{mm} > 109{,}25\ \text{mm}$✓

Use $Z = 109{,}25\ \text{mm}$ (least)✓ (3)

3.5 Tension reinforcement:

$$As = \frac{BM}{0{,}87\ fy\ Z}$$ CL 4.3.3.4.1

$$As = \frac{8{,}46 \times 10^6}{0{,}87 \times 250 \times 109{,}25}$$

$As = 356\ mm^2$✓

Use R12 bars @ 300 c/c ($As = 377\ mm^2$)✓ (3)

3.6 Secondary reinforcement:

$$\frac{100\ AS}{AC} = 0{,}24$$✓ Table 23 CL 4.11.4.3

$$As = \frac{0{,}24 \times AC}{100}$$

$$As = \frac{0{,}24 \times 1\,350 \times 140}{100}$$✓

$As = 405\ mm^2$

Use R10 bars @ 175 c/c ($As = 449\ mm^2$)✓ (3)

[23]

QUESTION 4

Throat thickness $= \sin 45° \times 8$ mm✓
$= 5{,}66$ mm✓

Effective length:

$F = P \times A_{eff}$ $P = 130$ MPa 10.7.1.2

$F = P \times throat \times L$

$$L = \frac{F}{P \times throat}$$

$$L = \frac{177 \times 10^3}{130 \times 5{,}66}$$

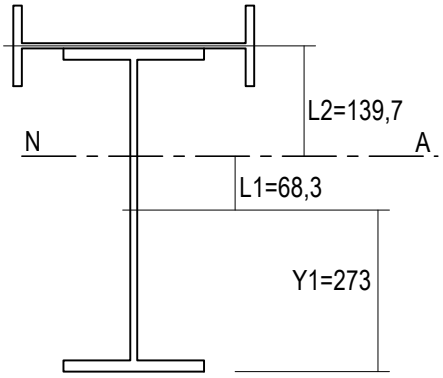
$L = 260{,}6$ mm✓ (130,3 mm per side)✓ **[6]**

QUESTION 5

I-section 406 × 178 × 67,2 kg/m:	I-section 305 × 102 × 32,8 kg/m:
$I_{xx} = 243,3 \times 10^{-6}\ m^4$	$I_{yy} = 1,941 \times 10^{-6}\ m^4$
Area $= 8,554 \times 10^{-3}\ m^2$	Area $= 4,183 \times 10^{-3}\ m^2$
$\frac{h}{2} = \frac{409,4}{2} = 204,7$ mm	$\frac{t1}{2} = \frac{6,6}{2} = 3,3$ mm

Total area = $8,554 \times 10^{-3}\ m^2 + 4,183 \times 10^{-3}\ m^2$

Total area = $12,737 \times 10^{-3}\ m^2$✓ (1)



Neutral axis using area moments from bottom:

$12,737 \times 10^{-3}\ m^2 \times Y1 = (8,554 \times 10^{-3}\ m^2 \times 0,2047) + (4,183 \times 10^{-3}\ m^2 \times 0,4127)$✓

$$Y_1 = \frac{1,751 \times 10^{-3}\ m^2 + 1,726 \times 10^{-3}\ m^2}{12,737 \times 10^{-3}\ m^2}$$✓✓

$$Y_1 = \frac{3,477 \times 10^{-3}\ m^2}{12,737 \times 10^{-3}\ m^2}$$✓

$Y_1 = 0,273$ m

$Y_1 = 273$ mm✓

$L1 = 273 - 204,7 = 68,3$ mm

$L2 = 204,7 + 3,3 - 68,3 = 139,7$ mm (5)

Second moment of area:

$I_{xx\ total} = (Ixx\ \ beam + \ al^2) + (Iyy\ \ beam + al^2)$

$I_{xx\ beam} = (243,3 \times 10^{-6} + 8,554 \times 10^{-3} \times 0,0683^2)$
$= (243,3 \times 10^{-6} + 39,9 \times 10^{-6})$
$= 283,2 \times 10^{-6}\ m^4$✓✓

$I_{yy\ beam} = (1,941 \times 10^{-6} + 4,138 \times 10^{-3} \times 0,1397^2)$
$= (1,941 \times 10^{-6} + 81,64 \times 10^{-6})$
$= 83,58 \times 10^{-6}\ m^4$✓✓

$I_{xx\ total} = 366,78 \times 10^{-6}\ m^4$✓✓ (6)

Bending moment maximum:

$\frac{M}{I} = \frac{f}{y}$ where $M = \frac{I \times f}{y}$

$BM_{max} = ¿\ \frac{366,78 \times 10^6 \times 162}{273}$✓✓

$BM_{max} = ¿\ 217,65\ kNm$✓ (3)

Central point load:

Self-weight = (67,2 kg/m + 32,8 kg/m) × 9,81 × 10^{-3} = 0,981 kN/m✓

$BM = \frac{W\ l^2}{8} + \frac{Wl}{4}$

$217,65 = \frac{0,981 \times 12^2}{8} + \frac{W \times 12}{4}$✓✓

$217,65 = 17,568 + 3W$

$W = \frac{217,65 - 17,568}{3}$✓

$W = 66,69\ kN$✓ (5)

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QUESTION 6

All references taken from SANS 10100-1 (2000)

Fcu = 30 MPa Table 2 (4.1.5.1)
Fy = 450 MPa Table 3 (4.1.5.2)

6.1 6.1.1 Nett area of the concrete:

$$\text{Nett area of the steel} = 4\left(\frac{\pi d^2}{4}\right) + 2\left(\frac{\pi d^2}{4}\right)\checkmark$$

$$= 4\left(\frac{\pi 20^2}{4}\right) + 2\left(\frac{\pi 16^2}{4}\right)\checkmark$$

$$= 1\,256{,}76 + 402$$

$$\text{Nett area of the steel} = 1\,658{,}76\ \text{mm}^2\checkmark$$

$$\text{Nett area of concrete} = (600 \times 200) - 1\,658{,}76\ \text{mm}^2$$
$$= 118\,341{,}24\ \text{mm}^2\checkmark \quad (4)$$

6.1.2 Axial load:

$N = 0{,}4\ f_{cu}\ A_c + 0{,}67\ f_y\ A_{sc}$ CL 4.7.4.3

$N = (0{,}4 \times 30 \times 118\,341{,}24) + (0{,}67 \times 450 \times 1\,658{,}76)\checkmark\checkmark$

$N = 1\,420\,094{,}8 + (500\,116{,}14)\checkmark$

$N = 1\,920{,}2\ \text{kN}\checkmark$ (4)

6.1.3 Diameter and spacing of binders:

Binders: ¼ of the smallest compression bar CL 4.11.4.5.1

¼ × 16 = 4 mm (not available)✓

Use: min R8 binders (R6 discontinued)

Spacing of binders:

12 × diameter of smallest compression bar
12 × 12 = 144 mm✓

Use spacing of 140 mm. (2)

6.2 Area of pad foundation:

$$\text{Area} = \frac{\Sigma \ of \ \ downward \ \ loads}{Upward \ \ soil \ \ pressure}$$

$$\text{Area} = \frac{3\,400 \ kN \ + \ 600 \ kN \ + \ 85 \ kN}{220 \ kN/m^2} \checkmark\checkmark$$

$$\text{Area} = 18,568 \text{ m}^2 \checkmark$$

$$\text{Size of foundation} = \sqrt{18{,}568 \ m^2}$$

$$4{,}31 \text{ m} \times 4{,}31 \text{ m} \checkmark$$

Use base size of 4,5 m × 4,5 m.✓ (4)
[14]

QUESTION 7

7.1 **Ferrous metals**
- cast iron
- structural steel
- carbon steel (Any TWO)

Nonferrous metals
- copper
- tin
- zinc
- lead
- aluminium (Any TWO)

(2 + 2) (4)

7.2
- Metal base plate
- Galvanised iron mould
- Tamping rod with diameter 12 mm × 600 mm (3)

[7]

TOTAL: 100

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